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Silicon Errata for the CYT2B9 Series Rev. C

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This document describes the errata for CYT2B9 Rev. C Series of devices. Details include trigger conditions, scope of impact, available workarounds, and applicable silicon revisions. Contact your local Infineon Sales representative for further questions.

Part Numbers Affected

Part Number
All CYT2B9 Rev. C parts

CYT2B9 Rev. C Qualification Status

Product Status: Engineering/Production Samples

CYT2B9 Rev. C Errata Summary

This table defines the errata applicable to CYT2B9 Rev. C Series of devices.

Items	Part Number	Silicon Revision	Fix Status
Errata ID 67 ConfigureFmInterrupt API assumes a parameter with 8 bytes boundary, but actual boundary is 4 bytes	CYT2B93CACES CYT2B94CACES CYT2B95CACES CYT2B97CACES CYT2B98CACES CYT2B93BACQ0AZSGS CYT2B93BACQ0AZEGS CYT2B93CACQ0AZSGS CYT2B93CACQ0AZEGS CYT2B94BACQ0AZSGS CYT2B94BACQ0AZEGS CYT2B94CACQ0AZSGS CYT2B94CACQ0AZEGS CYT2B95BACQ0AZSGS CYT2B95BACQ0AZEGS CYT2B95CACQ0AZSGS CYT2B95CACQ0AZEGS CYT2B97BACQ0AZSGS CYT2B97BACQ0AZEGS CYT2B97CACQ0AZSGS CYT2B97CACQ0AZEGS CYT2B98BACQ0AZSGS CYT2B98BACQ0AZEGS CYT2B98CACQ0AZSGS CYT2B98CACQ0AZEGS	Rev. C	No silicon fix planned. Use workaround.
Errata ID 68 SMPU/MPU/PPU protection region size is limited to 2 GB			No silicon fix planned. Use workaround.
Errata ID 69 DirectExecute API may return error if called with arguments placed in SRAM			No silicon fix planned. Use workaround.
Errata ID 96 CAN FD RX FIFO top pointer feature does not function as expected			No silicon fix planned. Use workaround.
Errata ID 97 CAN FD debug message handling state machine not get reset to Idle state when CANFD_CH_CCCR.INIT is set			No silicon fix planned. Use workaround.
Errata ID 98 TPIU peripheral ID mismatch			No silicon fix planned.
Errata ID 124 Limitation of the memory hole in SCB register space			No silicon fix planned.
Errata ID 129 WDT service can be missed			No silicon fix planned. Use workaround.

Items	Part Number	Silicon Revision	Fix Status
Errata ID 130 Customer cannot calibrate ILO clock	CYT2B93CACES CYT2B94CACES CYT2B95CACES CYT2B97CACES CYT2B98CACES CYT2B93BACQ0AZSGS CYT2B93BACQ0AZEGS CYT2B93CACQ0AZSGS CYT2B93CACQ0AZEGS CYT2B94BACQ0AZSGS CYT2B94BACQ0AZEGS CYT2B94CACQ0AZSGS CYT2B94CACQ0AZEGS CYT2B95BACQ0AZSGS CYT2B95BACQ0AZEGS CYT2B95CACQ0AZSGS CYT2B95CACQ0AZEGS CYT2B97BACQ0AZSGS CYT2B97BACQ0AZEGS CYT2B97CACQ0AZSGS CYT2B97CACQ0AZEGS CYT2B98BACQ0AZSGS CYT2B98BACQ0AZEGS CYT2B98CACQ0AZSGS CYT2B98CACQ0AZEGS	Rev. C	Will be fixed in flash boot version 3.1.0.425 or later
Errata ID 133 RMA failure			Will be fixed in flash boot version 3.1.0.515 or later
Errata ID 136 Secret Device Keys cannot be used			Will be fixed in flash boot version 3.1.0.487 or later
Errata ID 137 Limitation of Work flash reading			No silicon fix planned. Use workaround.
Errata ID 138 ROM boot code clears to zero last 2 KB of SRAM			No silicon fix planned. Use workaround.
Errata ID 139 Limitation of programming SFlash Normal Access Restrictions (row 13)			No silicon fix planned. Use workaround.

67. ConfigureFmInterrupt API assumes a parameter with 8 bytes boundary, but actual boundary is 4 bytes

- **Problem Definition**
  STATUS_ADDR_PROTECTED will be returned if ConfigureFmInterrupt API is called with arguments stored in SRAM with 4 bytes boundary of available SRAM or protected boundary SRAM.
- **Parameters Affected**
  N/A
- **Trigger Condition(s)**
  Call ConfigureFmInterrupt API with arguments stored in SRAM at 4 bytes boundary of available SRAM or protected boundary of SRAM.
- **Scope of Impact**
  ConfigureFmInterrupt API will fail by returning STATUS_ADDR_PROTECTED error status when called with argument having 4 bytes boundary of available SRAM or protected boundary of SRAM.
- **Workaround**
  Allow 4 bytes margin (i.e. assume that API parameter size as 8 and store the arguments) for API (ConfigureFmInterrupt) parameter.
- **Fix Status**
  No silicon fix planned. Use workaround.

68. SMPU/MPU/PPU protection region size is limited to 2 GB

- **Problem Definition**
  If SMPU/MPU/PPU protection block size is configured for 4 GB (PROT_SMPU_SMPU_STRUCT_ATT0.REGION.SIZE = 31), then during protection check in SROM, the value of the internal uint32 variable will overflow (4G = 0x1 0000 0000). Therefore, SROM assumes the protection size equals zero, and no protection will be applied.
- **Parameters Affected**
  N/A
- **Trigger Condition(s)**
  Configure SMPU/MPU/PPU to protect with region size equal to 4 GB or the region size with value 31u.

- **Scope of Impact**
  If SMPU/MPU/PPU is configured to protect region size of 4 GB, then SROM software does not apply any protection as per the request.
- **Workaround**
  Use two protection blocks of region size equal to 2 GB if 4 GB region size protection is required.
- **Fix Status**
  No silicon fix planned. Use workaround.

69. DirectExecute API may return error if called with arguments placed in SRAM memory

- **Problem Definition**
  If DirectExecute API is called in master PC (other than PC0 or PC1) with arguments in SRAM_SCRATCH_ADDR, then the API will return STATUS_ADDR_PROTECTED status.
- **Parameters Affected**
  N/A
- **Trigger Condition(s)**
  Call DirectExecute API with arguments in SRAM_SCRATCH_ADDR and master PC configured > 1.
- **Scope of Impact**
  DirectExecute API, if called with master PC configured > 1 and arguments in SRAM_SCRATCH_ADDR, the API will return STATUS_ADDR_PROTECTED.
- **Workaround**
  Call DirectExecute API with master PC0 or PC1, if arguments are stored in SRAM memory.
- **Fix Status**
  No silicon fix planned. Use workaround.

96. CAN FD RX FIFO top pointer feature does not function as expected

- **Problem Definition**
  RX FIFO top pointer function calculates the address for received messages in Message RAM by hardware. This address should be re-start back from the start address after reading all messages of RX FIFO n size (n: 0 or 1). However, the address does not re-start back from the start address when RX FIFO n size is set to 1 (CANFD_CH_RXFnC.FnS = 0x01). This results in CPU/DMA to read messages from the wrong address in Message RAM.
- **Parameters Affected**
  N/A
- **Trigger Condition(s)**
  RX FIFO top pointer function is used when RX FIFO n size set to 1 element (CANFD_CH_RXFnC.FnS = 0x01).
- **Scope of Impact**
  Received message cannot be correctly read by using RX FIFO top pointer function, when RX FIFO n size set to 1 element.
- **Workaround**
  Any of the following.
  1) Set RX FIFO n size to 2 or more when using RX FIFO top pointer function.
  2) Do not use RX FIFO top pointer function when RX FIFO n size set to 1 element. Instead of RX FIFO top pointer, read received messages from the Message RAM directly.
- **Fix Status**
  No silicon fix planned. Use workaround.

97. CAN FD debug message handling state machine not get reset to Idle state when CANFD_CH_CCCR.INIT is set

- **Problem Definition**
  If either CANFD_CH_CCCR.INIT bit is set by the Host or when the M_TTCAN module enters BusOff state, the debug message handling state machine stays in its current state instead of being reset to Idle state. Configuring the bit CANFD_CH_CCCR.CCE does not change CANFD_CH_RXF1S.DMS.
- **Parameters Affected**
  N/A

- **Trigger Condition(s)**
  Either CANFD_CH_CCCR.INIT bit is set by the Host or when the M_TTCAN module enters BusOff state.
- **Scope of Impact**
  The errata is limited to the use case when the Debug on CAN functionality is active. Normal operation of CAN module is not affected, in which case the debug message handling state machine always remains in Idle state. In the described use case, the debug message handling state machine is stopped and remains in the current state signaled by the bit CANFD_CH_RXF1S.DMS. In case CANFD_CH_RXF1S.DMS is set to 0b11, DMA request remains active.
  Bosch classifies this as non-critical error with low severity, there is no fix for the IP. Bosch recommends the workaround listed also here.
- **Workaround**
  In case the debug message handling state machine has stopped while CANFD_CH_RXF1S.DMS is 0b01 or 0b10, it can be reset to Idle state by hardware reset or by reception of debug messages after CANFD_CH_CCCR.INIT is reset to zero.
- **Fix Status**
  No silicon fix planned. Use workaround.

98. TPIU peripheral ID mismatch

- **Problem Definition**
  TPIU peripheral ID indicates that it is M3-TPIU instead of M4-TPIU.
- **Parameters Affected**
  N/A
- **Trigger Condition(s)**
  When debugger reads PID registers for component identification.
- **Scope of Impact**
  The debuggers read the TPIU as M3-TPIU and no other impact other than this.
- **Workaround**
  No specific workaround required. Debuggers can use trace features.
  Fix Status
  No silicon fix planned.

124. Limitation of the memory hole in SCB register space

- **Problem Definition**
  The memory hole [offset address: 0x1000 to 0xFFFF] inside SCB register space is not aligned to the below defined spec. The offset address bits [15:12] are ignored and treated as 4'b0000, so write/read access to offset address [0x1000 to 0xFFFF], will actually happen to [0x0000 to 0x0FFF].
  - Access to address gaps in mapped memory space: writes are ignored and any read returns a zero.
- **Parameters Affected**
  N/A
- **Trigger Condition(s)**
  Access to the memory hole [offset address: 0x1000 to 0xFFFF] in SCB register space.
- **Scope of Impact**
  The memory hole [offset address: 0x1000 to 0xFFFF] in SCB register space is not aligned to other IP registers.
- **Workaround**
  Do not access to the memory hole [offset address: 0x1000 to 0xFFFF] in SCB register space.
  Fix Status
  No silicon fix planned.

129. WDT service can be missed

- **Problem Definition**
  If WDT service happens within 4 ILO clock cycles before DeepSleep entry, it clears the counter but does not fully complete an internal handshake. A service after DeepSleep wakeup may then be missed if it occurs less than 2 ILO clock cycles after the processor resumes clocking. After this time, the internal handshake is complete and servicing works normally.

- **Parameters Affected**
  N/A
- **Trigger Condition(s)**
  Service WDT within 4 ILO clock cycles before DeepSleep entry and within 2 ILO clock cycles of processor clock resuming.
- **Scope of Impact**
  WDT service after DeepSleep wakeup may be ignored and WDT continues counting.
  This can cause unintended WARN_ACTION or UPPER_ACTION, including interrupt, fault, and/or reset.
- **Workaround**
  Wait 130 µs or more after DeepSleep wakeup. (e.g., To measure 130 µs, software can read WDT_CNT register at wake up and make sure that WDT_CNT was incremented of 4 units before servicing WDT.) Afterwards, write '1' to WDT service (WDT_SERVICE.SERVICE) after waiting until WDT service (WDT_SERVICE.SERVICE) reads '0'.
- **Fix Status**
  No silicon fix planned. Use workaround.

130. Customer cannot calibrate ILO clock

- **Problem Definition**
  ILO calibration registers (CLK_TRIM_ILO0/1_CTL) are part of SRSS secure registers. Therefore, user software cannot write to them directly. Note that user software can write them directly on the following ES revision devices because their SRSS secure registers are un-protected.
  - Rev. A
- **Parameters Affected**
  N/A
- **Trigger Condition(s)**
  Write attempt to ILO calibration registers (CLK_TRIM_ILO0/1_CTL).
- **Scope of Impact**
  User software cannot write ILO calibration registers (CLK_TRIM_ILO0/1_CTL).
- **Workaround**
  None.
  We will prepare new flash boot version (3.1.0.425 or later) to enable access to ILO calibration.
- **Fix Status**
  Will be fixed in silicon Rev. C (CS).

133. RMA failure

- **Problem Definition**
  TransitionToRMA and OpenRMA always fail due to hard fault resulting from access to protected memory (User SRAM except first 2 KB of SRAM0).
- **Parameters Affected**
  N/A
- **Trigger Condition(s)**
  When using TransitionToRMA or OpenRMA
- **Scope of Impact**
  TransitionToRMA and OpenRMA always fail.
- **Workaround**
  None.
- **Fix Status**
  We will prepare new flash boot version (3.1.0.515 or later) to fix this erratum.
  Currently, first 2 KB of SRAM0 is reserved. However, additional 2 KB of SRAM0 will be used only for the RMA case.

136. Secret Device Keys cannot be used

- **Problem Definition**
  Secret Device Keys stored in eFuses can only be accessed by the crypto hardware module. Before the

crypto module read the secret device keys, the eFuse read operation has to be enabled. Current protection configuration of device prevents any software from enabling the read operation. Attempt to load the secret device keys into the crypto module will result in all-‘0’ data being loaded as secret key. The failure is unascertainable to the user since no error is reported.

- **Parameters Affected**
  N/A
- **Trigger Condition(s)**
  Reading the secret device keys from eFuse
- **Scope of Impact**
  Secret device keys cannot be read from eFuse. All-‘0’ data will be used by the crypto module instead.
- **Workaround**
  None. Do not use the secret device key feature.
- **Fix Status**
  We will prepare new flash boot version (3.1.0.487 or later) to enable eFuse read. And we will also prepare the HSM performance library which supports to read the secret device keys.

137. Limitation of work flash reading

- **Problem Definition**
  Work flash can be read via different CPU cores but only one CPU core is assigned for non-correctable ECC error handling.
- **Parameters Affected**
  N/A
- **Trigger Condition(s)**
  Reading work flash via CPU core (CM0+, CM4) and ECC fault interrupt routed to two CPU cores.
- **Scope of Impact**
  Only one CPU core can be assigned for non-correctable ECC error handling.
- **Workaround**
  Any of the following:
  Option A (Recommended solution)
  Set each CPU core to use a separate DMA (M-DMA or P-DMA) channel to read work flash. If non-correctable ECC error occurs, the DMA transaction get aborted and respective CPU core gets the interrupt to manage the non-correctable ECC error.

  Option B[1]
  Set non-correctable ECC error handling to reset. This way no one CPU core needs to manage the non-correctable ECC error handling.

  Note [1]: Not recommended to use for EEPROM emulation. EEPROM emulation needs to cope with aborted write/erase. In such a scenario, option B leads to deadlock in endless resets. However, option B can be used if work flash update is not intended in the field.

  Option C[2]
  Assign one CPU core for non-correctable ECC error handling and that core informs the error to the other core which caused the error, but it takes time.

  Note [2]: Not recommended to use with MCAL, since the inter-core communication is too slow.

- **Fix Status**
  No silicon fix planned. Use workaround.
  Infineon FLS and FEE driver will be updated with workaround A.
  TRM will be updated for this limitation.

138. ROM boot code clears to zero last 2 KB of SRAM

- **Problem Definition**
  ROM boot code clears to zero last 2 KB of SRAM. This region is available to the user after boot. However, data retention across resets is not guaranteed in this area.
- **Parameters Affected**
  N/A
- **Trigger Condition(s)**
  After ROM boot
- **Scope of Impact**
  Data retention across resets is not guaranteed in last 2 KB of SRAM.
- **Workaround**
  Do not use last 2 KB of SRAM for data retention.
- **Fix Status**
  No silicon fix planned. Use workaround.
  TRM will be updated for this limitation.

139. Limitation of programming SFlash Normal Access Restrictions (row 13)

- **Problem Definition**
  CM0+ cache is not disabled when programming SFlash Normal Access Restrictions (row 13) by WriteRow SROM API. Occasionally, writing to SFlash Normal Access Restrictions (row 13) may return error status “0xF00000A4” (ProgramRow is invoked on unerased cells or blank check fails).
- **Parameters Affected**
  N/A
- **Trigger Condition(s)**
  WriteRow SROM API is called on Normal Access Restrictions (row 13).
- **Scope of Impact**
  Error status – 0xF00000A4 “ProgramRow is invoked on unerased cells or blank check fails” is returned.
- **Workaround**
  Disable CM0+ cache before call to WriteRow (to SFlash row 13) and enabling the cache back after the SROM API execution. Following sequence could be a recommended sequence:
  1) FLASHC_CM0_CA_CTL0.CA_EN = 0; // Disable the CM0+ cache
  2) Trigger WriteRow SROM API on NAR (row 13)
  3) WriteRow successful
  4) FLASHC_CM0_CA_CTL0.CA_EN = 1; // Enable the CM0+ cache
- **Fix Status**
  No silicon fix planned. Use workaround.
  TRM will be updated for this limitation.

Revision history

Document revision	Date of release	Description of Change
1.0	2019-12-19	Initial release
1.1	2020-01-22	Updated the description in errata ID 129. Added new flash boot version to workaround of errata ID 130.
1.2	2020-06-17	Added errata ID 133.
1.3	2020-08-28	Updated the 'Fix Status' of errata ID 133. Added errata ID 136 to 139.

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